AP CALCULUS AB

Unit 1 • Self Check 1.3 and 1.4 • Lessons 1-5 to 1-9 • Answer key

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Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Which function must have an absolute minimum on the interval?

(A) $\frac{1}{x}$ on $[1,3]$

(B) $\frac{1}{x}$ on $(0,3]$

(C) x on $(0,1)$

(D) $\ln x$ on $(0,1]$

Solution

Answer: (A).

Question 2

NO CALCULATOR

For $f(x) = x^2$, $\frac{f(1+h)-f(1)}{h} =$

(A) 2

(B) $2+h$

(C) $1+h$

(D) h

Solution

Answer: (B).

Question 3

NO CALCULATOR

For $g(x) = x^3$ on $[0,3]$, the MVT value is $c =$

(A) 1.5

(B) 3

(C) $\sqrt{3}$

(D) $\sqrt[3]{9}$

Solution

Answer: (C).

Question 4 (Free response) NO CALCULATOR

Use the definition to find $f'(x)$ for $f(x) = x^2 - 3x$, write the tangent line at $x = 4$, and find the c given by the MVT on $[0, 4]$.

Solution
$f'(x) = 2x - 3$; $y - 4 = 5(x - 4)$; secant slope $\frac{4-0}{4} = 1 = 2c - 3$, $c = 2$.